HyProC - Manual

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Getting started

HyProC makes use of SRIM (Stopping and Range of Ions in Matter) [1] to calculate stopping powers. It thus has to be pre-installed for the code to work. Its path in the settings.json file ("SRIM_path" parameter).

From the HyProC folder, run the UI.py file to launch the code.

Button behaviours

Import excitation curve	Loads an excitation curve. To get data, the code will read the first tab of file and will look for the "Energie (keV)", "Ng/µC" and "Incertitude" headers on the second row. To change that, simply edit the settings.json file (no need to close the program). File format: Excel (.xlsx, .xls, .csv) Having the file opened in Excel will cause an error.
Run calculation	Simulates an excitation curve, i.e. calculates the yield for each energy value of the loaded excitation curve. Thus, an excitation curve has to be loaded for the function to run.
Extract H profile	Plots the hydrogen depth profile based on the current target.
Load target	Self-explanatory.
Save target	File format: .json
Save simulated curve data	Saves the data of the simulated excitation curve in a two-column tab-separated .txt file.

(Standard) Load	Self-explanatory. Since a standard should be uniform, it should contain only one layer. Thus when loading a multi-layer target as standard, only its first layer will be used. File format: .json. When saving a multi-layer standard, the extra layers are NOT removed.
(Standard) Save	
Scroll Up/Down	Self-explanatory.
Add/Remove Layer Move Up/Down Duplicate Layer	Self-explanatory.
Add/Remove element	Self-explanatory.
Isolate & Normalize	Rescales atomic percentages in a layer so they add up to 100%, while keeping the selected element's percentage fixed. It proportionally adjusts the other elements' values to maintain balance. This is especially useful when incorporating an element into a compound (e.g. hydrogen in SiO_2). Define the compound with the correct atomic concentrations, add the new element with the desired concentration, then hit the button.

Options

Beam spread

An ion beam is rarely mono-energetic, even before it reaches the target. To reflect this, HyProC assumes a gaussian energy distribution of the beam. The user can specify the standard deviation of this gaussian distribution in the text field next to "Beam spread StDev (keV)". If set to 0, the beam is considered mono-energetic.

Doppler

A doppler effect causes a gaussian broadening of the reaction's resonance [2]. This can be taken into account or not using the tickbox. The standard deviation of this gaussian broadening is calculated by the composition of the layer in which

Broadening data

Broadening data can be saved by ticking the "Save broadening data" box. When calculations are launched with this option on, folders will be created using this architecture:

```
C:/HyProC/
└── Session/
    └── Run/
        └── Datapoint/
            ├── Beam.txt
            ├── Stragg.txt
            ├── Doppler.txt
            ├── Total_Gauss.txt
            ├── xsec.txt
            ├── Total_Broadening.txt
            └── Total_Broadening_TFU.txt
```

Total_Gauss represents the convolution of all gaussian broadening (Beam, Straggling and Doppler). Convoluting this curve with the cross section of the reaction (xsec, modelled as a lorentz curve) gives a Voigt profile representing the Total_Broadening. Total_Broadening_TFU represents the same data with a different x-axis (areal density in TFU instead energy).

A Session folder is created when broadening data are saved for the first time after the UI is started. A Run folder is created whenever a calculation is made (with the save broadening data on). For each incident energy, a datapoint folder is created.

The broadening data can be visualised using the broadplotter.py script (recommended) or through a third-party software (OriginLab, ...).

Chi-square test

Once a simulated excitation curve is generated, a chi-square test is made to assess how well it fits the loaded experimental curve. The value displayed on the "Chi-square history" graph is obtained with this formula:

$$\chi^2 = \frac{\sum_{i=0}^{N} (y_{\mathrm{exp},i} - y_{\mathrm{sim},i})^2}{N}$$

References

[1] James F Ziegler, Matthias D Ziegler, and Jochen P Biersack. "SRIM–The stopping and range of ions in matter (2010)". In: *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* 268.11-12 (2010), pp. 1818–1823.

[2] KM Horn and WA Lanford. "Observation of the bond-dependent Doppler broadening of the p (15N, $\alpha\gamma$) 12C nuclear reaction". In: *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* 29.4 (1988), pp. 609–620.